Beyond Content: Temporal Dynamics of Long-Form Human-LLM Interaction

Author: Kristina Baldovino
Affiliation: Independent Researcher
ORCID iD: 0009-0007-0457-4342
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Abstract

Most analyses of Human-LLM conversation focus on semantic content, task performance, or turn-level behavior. Less attention has been paid to the temporal structure of long-form interactions as dynamical processes unfolding over hundreds or thousands of turns. In this exploratory study, we examine whether extended Human-LLM conversations exhibit repeatable temporal patterns when semantic content is removed.

We analyzed a corpus of 25 long-form Human-LLM conversations involving a single human participant with a consistent interaction style. Each conversation is transformed into a normalized time series representation based on response magnitude, enabling comparison across varying lengths and topics. Using rolling window smoothing, temporal resampling, and unsupervised clustering, we observe the emergence of multiple distinct interaction regimes characterized by stable rhythm envelopes.

To test robustness, we apply perturbation analyses, including additive stochastic perturbation proportional to series variance, and local permutation of temporal order (surrogate sequence testing). Observed regime structure remains largely intact under noise but degrades when temporal order is disrupted, including sensitivity to sequence rather than static statistics. These findings suggest that long-form Human-LLM interactions can occupy a small number of stable, path-dependent regimes under fixed interactional boundary conditions.

This work is descriptive and conditional in scope. Results are not claimed to generalize across users, models, or interaction styles. Instead, we argue that temporal dynamics constitute a meaningful and underexplored dimension of Human-LLM interaction, warranting further study alongside semantic and task-based analysis.

1.0 Methods

1. 1 Data

This study analyzes a corpus of 25 long-form Human-LLM conversations involving a single human participant. Conversations span a range of topics and lengths, each consisting of several hundred to over one thousand turns. All conversations were conducted using LLMs under standard conversational settings, without external memory, roleplay, romantic framing, or explicit identity assignment.

The human participant maintained a consistent interaction style across conversations, characterized by extended inquiry, tolerance for ambiguity, and avoidance of narrative closure. The consistency allows the human side of the interaction to be treated as a fixed boundary condition for the purposes of exploratory analysis.

Conversations were stored as transcripts and converted into structured, turn-level CSV representations using a lightweight preprocessing pipeline. For each turn, we extracted a scalar proxy for response magnitude (e.g., estimated token count). No semantic embeddings, sentiment labels, or content-based features were used in this analysis.

1.2 Time Series Representation

For each conversation, turns were ordered sequentially and mapped to a univariate time series representing response magnitude as a function of turn index. Because conversations varied substantially in length, direct comparison required normalization across time and scale.

To emphasize interactional envelope rather than local fluctuation, each time series was smoothed using a rolling mean with a fixed window size. The resulting smoothed series was then resampled to a fixed number of points using linear interpolation, mapping each conversation onto a normalized temporal axis expressed as a percentage of total conversation length. Each resampled series was z-score normalized to remove absolute magnitude differences and focus analysis on relative temporal structure.

Response magnitude was used as a scalar proxy for interactional intensity, reflecting the degree of elaboration produced by the system at each turn. No claims are made regarding internal cognitive states or uncertainty.

1.3 Similarity and Clustering

Normalized time series representations were compared using standard distance metrics (cosine and Euclidean distance). Unsupervised hierarchical clustering was performed on the resulting distance matrix using Ward linkage to identify groups of conversations with similar temporal structure.

Cluster assignments were not pre-specified, and no topic or metadata labels were used. Cluster interpretation was based solely on the shape of the normalized temporal trajectories, and the corresponding cluster mean curves.

1.4 Perturbation Analysis

To test whether observed structure reflected genuine temporal dynamics rather than static statistical properties, two perturbation procedures were applied:

1. Magnitude Noise Injection:
   Random Gaussian noise proportional to each conversation's magnitude variance was added to the original time series prior to smoothing and normalization. This preserves temporal order while perturbing absolute values.

2. Temporal Scrambling:
   Conversation order was disrupted by locally scrambling contiguous windows of turns, preserving magnitude distribution while degrading sequential structure.

For each perturbation condition, the full normalization and clustering pipeline was re-run, and resulting cluster structure was compared qualitatively to the unperturbed case.

1.5 Scope and Limitations

This analysis is exploratory and conditional. Results are not claimed to generalize across users, models, or conversational contexts. The use of response magnitude as a proxy variable captures temporal rhythm but does not encode semantic or affective information.

The purpose of this study is not to propose a taxonomy of conversation types, but to demonstrate that temporal dynamics alone can reveal regimes under controlled boundary conditions.

Future work may examine whether similar temporal regimes emerge under different interactional boundary conditions, including variations in pacing, framing, and user intent. Further extensions may incorporate additional temporal features, speaker-specific trajectories, or quantitative regime stability metrics. The present work serves as an exploratory foundation for such investigations.

1.6 Reproducibility

All preprocessing, normalization, clustering, and perturbation steps were implemented using standard numerical and statistical tools. Parameter choices (e.g., smoothing window size, resampling length) were held constant across all conversations. Code and analysis artifacts are available in associated repositories.

2.0 Results

2.1 Emergence of Repeatable Temporal Regimes

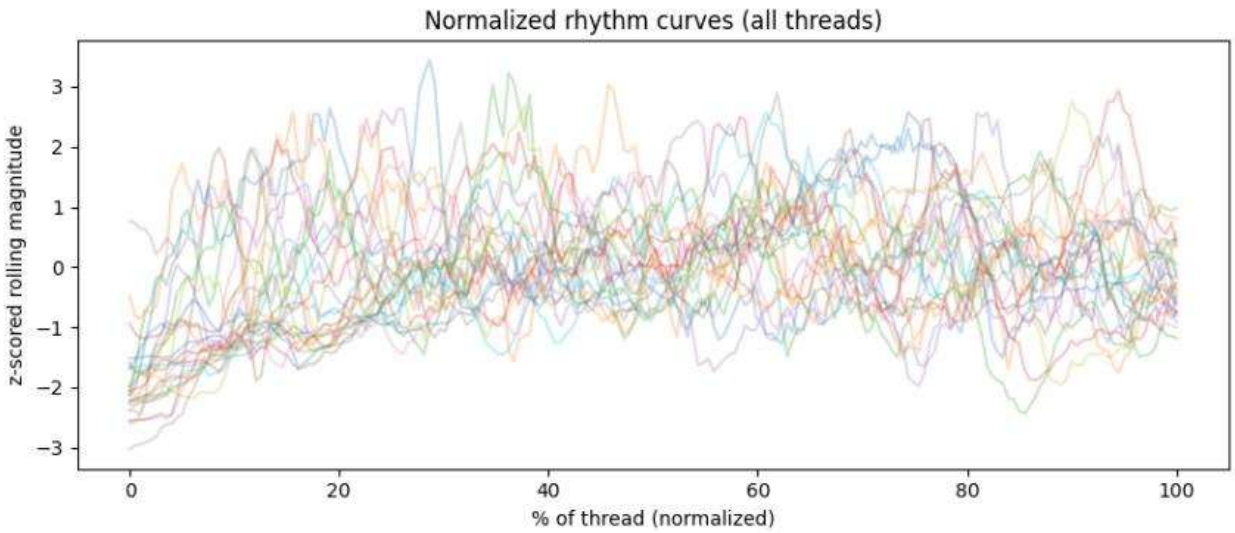

Fig. 1: Normalized rolling-mean trajectories for 25 long-form conversations, resampled to a fixed temporal axis. While individual trajectories vary locally, global envelope structure is preserved.

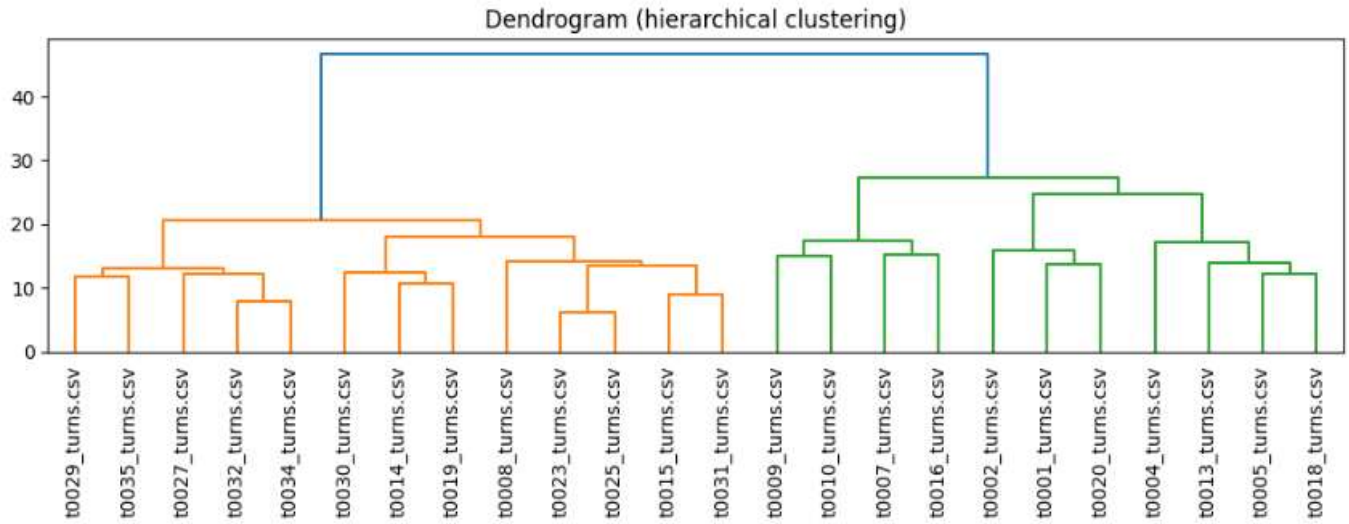

Fig. 2: Hierarchical clustering dendrogram of normalized conversation trajectories using Ward linkage. Clustering is based solely on temporal structure of response magnitude trajectories, without semantic or topical features. Distinct groupings indicate the presence of multiple interaction regimes within the analyzed corpus.

After normalizing and resampling all 25 long-form conversations, we observed that interaction trajectories did not collapse into noise or a single averaged profile. Instead, the normalized

time series representations exhibited visible structure, with multiple distinct rhythm envelopes emerging across conversations.

Unsupervised hierarchical clustering of the normalized trajectories revealed a small number of stable clusters. At the chosen clustering resolution, three primary interaction regimes were observed. These clusters were defined solely by temporal structure. No semantic, topical, or metadata features were used in the clustering process.

Within each cluster, individual conversation trajectories displayed similar envelope shapes, and the curves highlighted distinct temporal profiles. While individual trajectories varied locally, their global temporal organization remained consistent within clusters.

2.2 Robustness to Perturbation

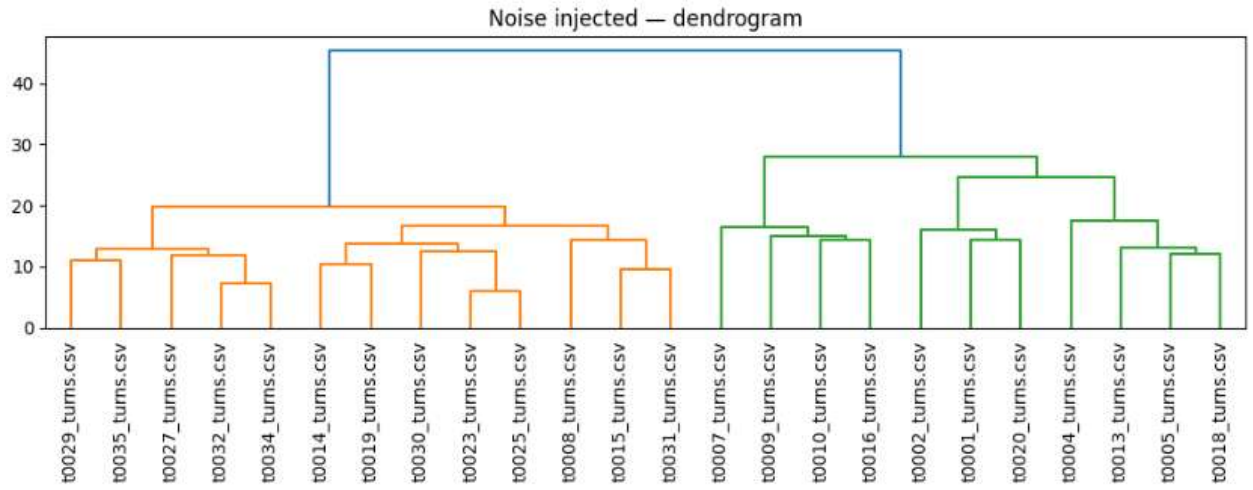

Fig. 3: Hierarchical clustering dendrogram after additive stochastic perturbation of response magnitude values. Gaussian noise proportional to each series' variance was applied prior to smoothing and normalization, preserving temporal order while perturbing values. Overall cluster structure remains largely intact, suggesting robustness to magnitude noise.

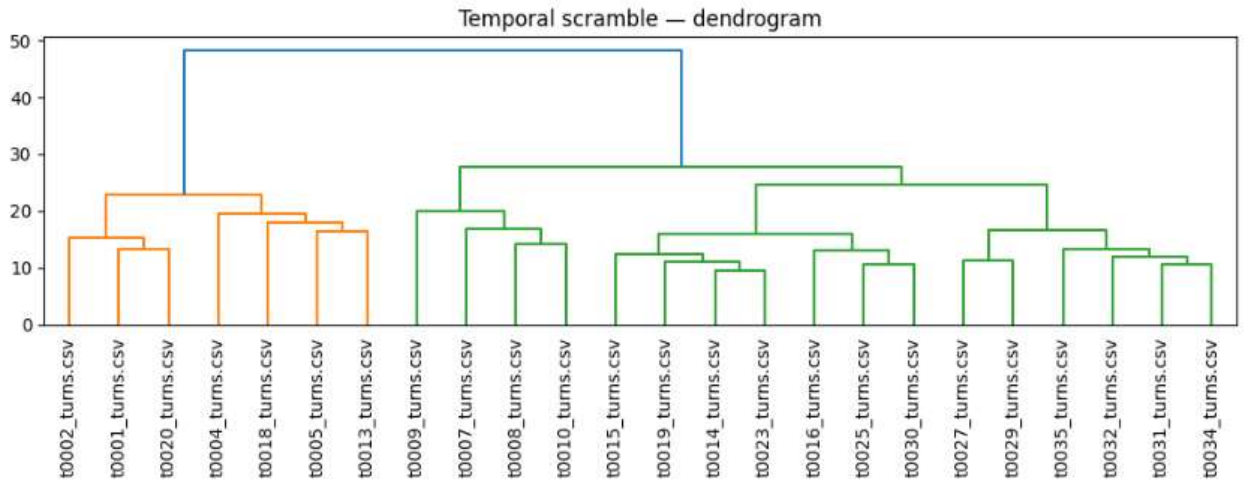

Fig. 4 Hierarchical clustering dendrogram following temporal scrambling via local permutation of turn order. This procedure preserves magnitude distributions while degrading sequential structure. Compared to the unperturbed case, cluster separation is reduced, indicating sensitivity of the observed regimes to temporal ordering.

To assess whether the observed regimes reflected genuine temporal dynamics rather than static statistical properties, perturbation analyses were performed.

When additive stochastic noise proportional to each series' variance was applied prior to normalization, the overall cluster structure what largely preserved. While local variation increased, conversations tended to remain grouped within their original clusters, indicating robustness to magnitude perturbation.

In contrast, when temporal order was disrupted through local permutation of turn sequences, the cluster structure degraded substantially. Dendogram separation became less distinct, and conversations that clustered together in the unperturbed case were no longer consistently grouped. This sensitivity to temporal scrambling suggests that the observed regimes depend on sequential organization rather than on static distributional features alone.

2.3 Scale Effects

Increasing the sample size from an initial exploratory subset to the full set of 25 conversations did not weaken the observed structure. Instead, cluster separation becomes more pronounced, and cluster mean profiles stabilized. This indicates that the identified regimes are not artifacts of small sample size or selective sampling within the analyzed corpus.

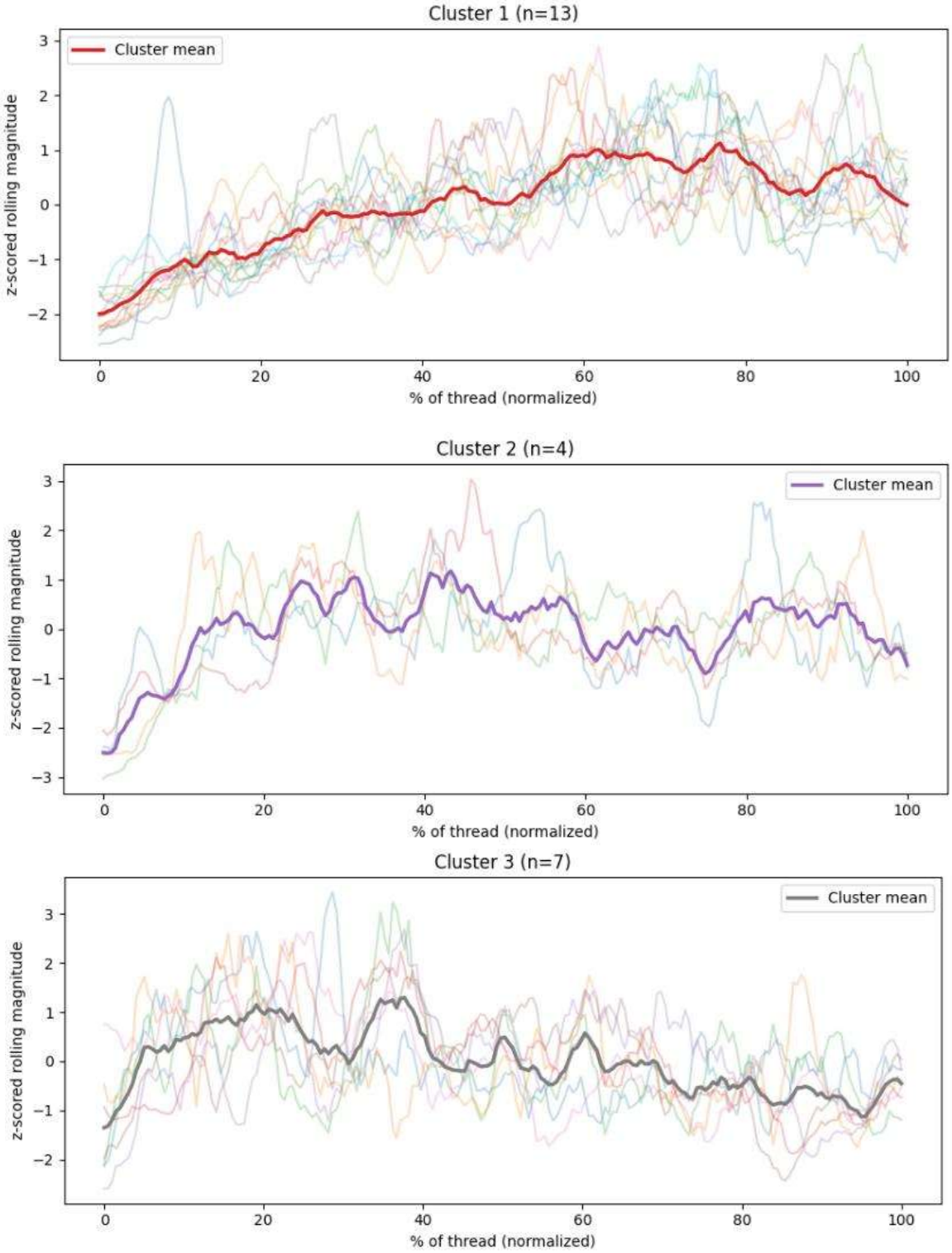

Fig. 5 Mean normalized temporal trajectories for each identified interaction regime. Faint lines represent individual conversation trajectories within a cluster; bold lines show cluster means. Regimes differ in overall temporal shape, including ramp-up, modulation, and late-stage settling. These profiles describe interaction dynamics rather than semantic or psychological categories.

3.0 Discussion

3.1 Interpretation of Temporal Regimes

The observed interaction regimes suggest that long-form Human-LLM conversations can exhibit repeatable temporal structure under fixed interactional boundary conditions. These regimes are characterized by distinct patterns of ramp-up, intensity modulation, and late-stage settling, independent of conversational topic or semantic content.

Importantly, these regimes do not imply categorical conversation “types” or psychological states. Rather, they represent dynamical patterns describing how interaction unfolds over time. The presence of multiple regimes indicates that conversational dynamics occupy a structured space with limited degrees of freedom, rather than an unbounded continuum.

3.2 Stability Without Narrative Closure

Unlike interaction regimes driven by strong narrative or emotional reinforcement (e.g., romantic or identity-framed interactions), the regimes observed here remain structurally open. Conversations frequently return toward baseline rather than converging on a single high-intensity state. This distinction suggests that temporal stability can arise from interaction dynamics alone, without requiring narrative closure or belief-driven stabilization.

3.3 Implications for Human-LLM Interaction Research

These findings highlight temporal dynamics as an underexplored dimension of Human-LLM interaction. Current evaluation approaches often emphasize turn-level correctness, semantic similarity, or task completion. The present analysis suggests that examining conversation as a trajectory may reveal structure invisible to content focused methods.

Understanding interaction regimes may inform future work on long-form dialogue stability, breakdown prediction, and interaction design. Such analysis complements rather than replaces semantic or task-based evaluation.

4.0 Acknowledgements

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5.0 Author Contributions & AI Assistance Disclosure

This manuscript presents original theoretical work and conceptual modeling by Kristina Baldovino. The author is solely responsible for the research design, analytical synthesis, and interpretive conclusions. Portions of the manuscript text (including grammar refinement and formatting) were prepared with assistance from an OpenAI language model, under the author's direct guidance. The model was used as a technical writing aid only. All intellectual content and editorial decisions remain the sole responsibility of the author.

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